Junzhe “Calvin” Yang

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EDUCATION

University of California, Irvine — Irvine, CA
Bachelor of Science, Computer Science and Engineering — *Expected Jun. 2030*
GPA: 4.00/4.00

SKILLS

Programming Languages: Java, Python, C/C++, JavaScript, HTML/CSS
Software: ANSYS Maxwell, Fusion 360, VS Code, EasyEDA, OpenRocket, NumPy, pytest, Git, GitHub Actions, CapCut, Microsoft Office, Adobe Photoshop, Claude, Codex

PROJECTS

Electromagnetic Accelerator (Coil Gun) — Jun. 2022 – Present
Individual Project
- Designed and built a two-stage coil gun, solving the challenge of precisely timing sequential electromagnetic coils to efficiently accelerate a projectile.
- Used ANSYS Maxwell 2D for electromagnetic field simulation and coil-timing optimization, Fusion 360 for mechanical design, 3D printing for fabrication, and Arduino for stage-control logic.
- Achieved approximately 11 m/s projectile velocity with a 54.64 g projectile using four 400 V, 330 µF capacitors.

Coilgun Timing Simulator — Aug. 2026 – Sep. 2026
Individual Project
- Built a Python simulator coupling capacitor-discharge circuit dynamics to projectile motion through a position-dependent inductance model, solving the resulting 5-variable coupled ODE system with a custom 4th-order Runge-Kutta integrator.
- Validated numerically rather than by fitting: matched the closed-form RLC solution, confirmed 4th-order convergence (16.0× error reduction per step halving), and enforced energy conservation to 6×10^{-14} relative across circuit, kinetic, and resistive terms.
- Automated 22 tests across 3 Python versions in GitHub Actions; quantified a 46.8% velocity loss to magnetic suckback and identified a projectile-placement change predicted to recover 37% of muzzle velocity.

Arduino Sensor & Control System — Jun. 2026 – Jul. 2026
Individual Project
- Built a real-time sensor monitoring system using an Arduino Uno R3, integrating an ultrasonic distance sensor and temperature/humidity sensor with an OLED display.
- Developed C/C++ firmware to collect sensor measurements and refresh the OLED display at 50 ms intervals.
- Implemented I2C communication between the Arduino, sensors, and OLED display.

EXPERIENCE

UCI Master of Engineering Capstone Project Assistant — Jan. 2025 – Jun. 2025
Gaudi Technology — *Irvine, CA*
- Revised 3D CAD models in Fusion 360 for printable concrete structures.
- Assisted with fabrication and preparation of concrete molds used for ASTM material strength testing.
- Collaborated with 4 MoE student researchers during prototype development and testing.

ACTIVITIES & LEADERSHIP

American Rocketry Challenge — Nov. 2025 - Mar. 2026
R&D and Simulation Lead
- Improved simulated flight accuracy by 23.2% through iterative OpenRocket analysis.
- Collaborated with a 7-member engineering team to design and build a 25.6-inch competition rocket.
- Completed two competition qualification flights, achieving an official score of 139.8 total.

YouTube (youtube.com/@SomeHeresyGaming) — Jan. 2026 - Present
Content Creator
- Built and grew a YouTube channel to 50,000+ subscribers and 150M+ total views.